# Study Design and Evaluation Practices in Clinical Artificial Intelligence: A Large-Scale Analysis of the Biomedical Informatics Literature

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## Summary

**Background**
Artificial intelligence is increasingly used in clinical and biomedical informatics. However, the structure of the literature—particularly study design and evaluation—remains uneven.

**Methods**
PubMed was searched for studies involving machine learning and biomedical informatics between 2016 and 2025, generating 4,650 records. Of these, 972 studies with available abstracts were included. Records without abstracts were excluded due to insufficient information for consistent extraction. The study design and evaluation methods were derived from titles and abstracts, and the temporal trends and relationships between study design and evaluation were examined.

**Findings**
The literature is dominated by methods for model development and application-focused studies, with limited representation of clinical or observational designs. Evaluation practices vary between study types and are not consistently applied. The volume of publications increases steadily through 2025. Work framed as decision support is more prevalent than explicitly reasoning-based approaches.

**Interpretation**
Methodological development in clinical AI has advanced rapidly, but alignment with the design and evaluation of clinical studies remains incomplete. Greater consistency in both will be required for clinical translation.

## Research in Context

**Evidence prior to this study**
Previous work has focused on individual applications of machine learning in healthcare. There has been less attention to how studies are structured and evaluated in the literature as a whole.

**Added value of this study**

This study provides a structural analysis of clinical AI research, showing how study design, evaluation practices, and conceptual framework vary across the field and over time.

**Implications of all the available evidence**

Clinical AI is advancing rapidly, but alignment with clinical study design and reasoning remains incomplete. Addressing this gap will be important for translation into routine clinical use.

# 1 Introduction

Artificial intelligence now plays a pivotal role in biomedical informatics and clinical research. Machine learning models are extensively used on clinical datasets, such as electronic health records and medical imaging. However, despite this rapid expansion, the overall organization of the literature is still not well characterized. Many publications focus solely on reporting model performance. Far less emphasis has been placed on how studies are designed and assessed at scale. This gap makes it difficult to gauge the field's maturity and its readiness for clinical translation. The present study analyzes the structure of the clinical AI literature, with particular attention to study design, evaluation methodologies, and how these have changed over time.

# 2 Methods

PubMed was searched for studies on artificial intelligence in biomedical informatics published between 2016 and 2025, using a combination of MeSH terms and free-text keywords to include both indexed and non-indexed entries, resulting in 4,650 initial records. Titles, abstracts, and related metadata were then extracted and harmonized. Records lacking abstracts were removed because they did not provide sufficient information for uniform analysis, leaving a final corpus of 972 studies.

The retrieval procedure relied on rule-based operations defined at the query level, while the subsequent analysis followed a model-driven approach. Abstracts were encoded using transformer-based embedding models to represent semantic relationships across the entire corpus. The resulting embeddings supported similarity analysis, clustering, and the discovery of underlying thematic patterns. A task-tailored large language model was created to assist with study design classification and the interpretation of ambiguous instances. The model was used in a restricted context, and its outputs were checked against predefined criteria to ensure consistency across the data set. In practice, incorporating the language model enhanced the reliability of classifications, especially for cases where embedding-based approaches alone were insufficient to resolve uncertainty. Evaluation metrics (such as AUC, accuracy, F1 score, and cross-validation) were detected through targeted pattern matching performed on the abstract text. Concept-level features—such as mentions of clinical reasoning and decision support—were detected using broadened lexical patterns derived from embedding-based similarity, enabling the identification of both explicit and implicit occurrences.

The final dataset was analyzed to characterize the distribution of study designs and evaluation methods, to examine how these aspects evolved over time, and to explore how study design was associated with evaluation practices. We used cross-tabulation to summarize the co-occurrence of different study design categories and evaluation methods. Temporal analyses were restricted to publications up to and including 2025, and concept-level trends were normalized by total publication counts to account for growth in the overall literature.

# 3 Results

## 3.1 Study Design

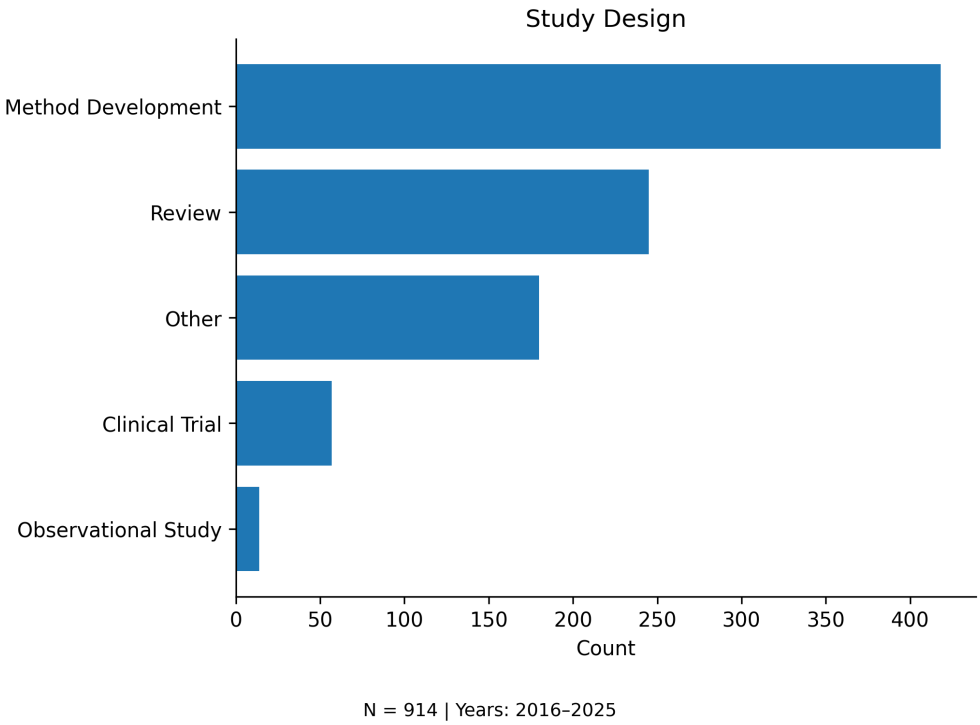


Figure 1: Distribution of study design categories.

Most publications focus on model development, whereas comparatively fewer studies address their evaluation in clinical practice using clinical or observational data.

### 3.2 Evaluation Methods

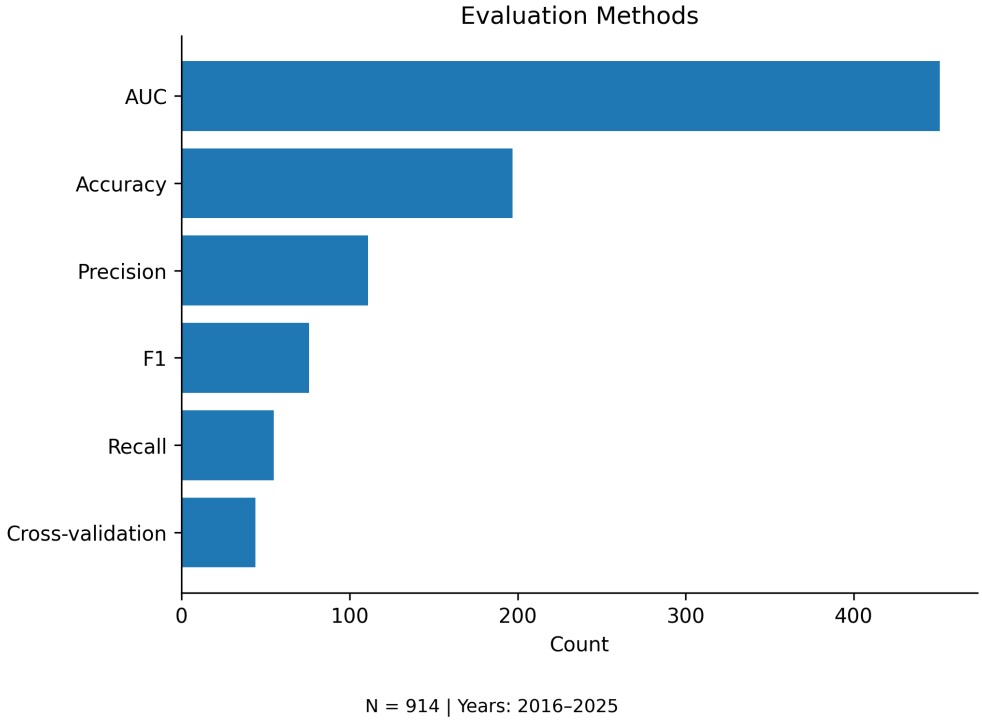


Figure 2: Distribution of evaluation methods.

Evaluation approaches differ across studies. While metrics like AUC, accuracy, and cross-validation are frequently reported, their application is not uniform across different study designs.

### 3.3 Temporal Trends

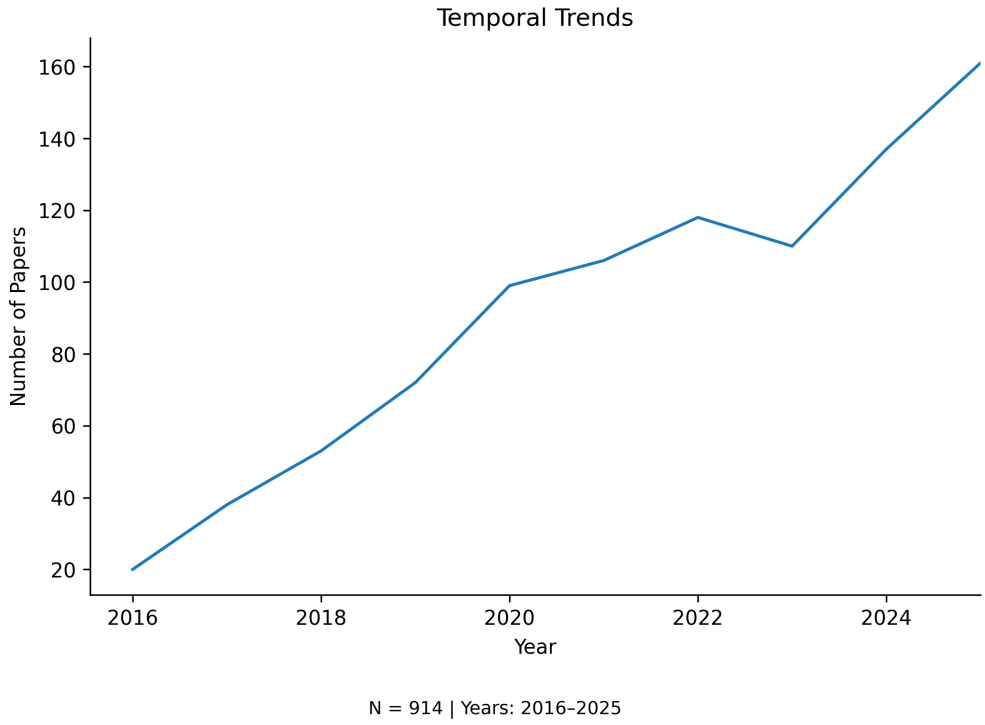


Figure 3: Temporal growth of publications through 2025.

The number of publications continues to grow consistently until 2025, primarily fueled by modeling and application-oriented research.

### 3.4 Study Design and Evaluation

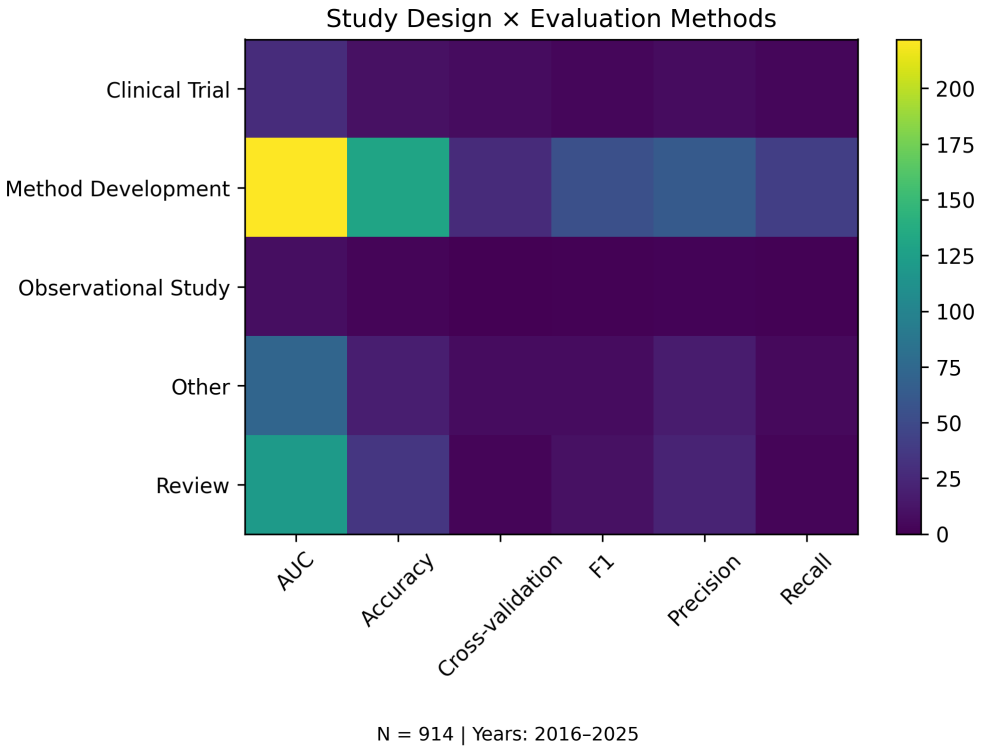


Figure 4: Study design versus evaluation methods.

Evaluation approaches vary across different study types. Method-oriented studies tend to rely more consistently on established metrics, whereas the remaining categories display greater diversity in how they are evaluated.

### 3.5 Clinical Reasoning vs Decision Support

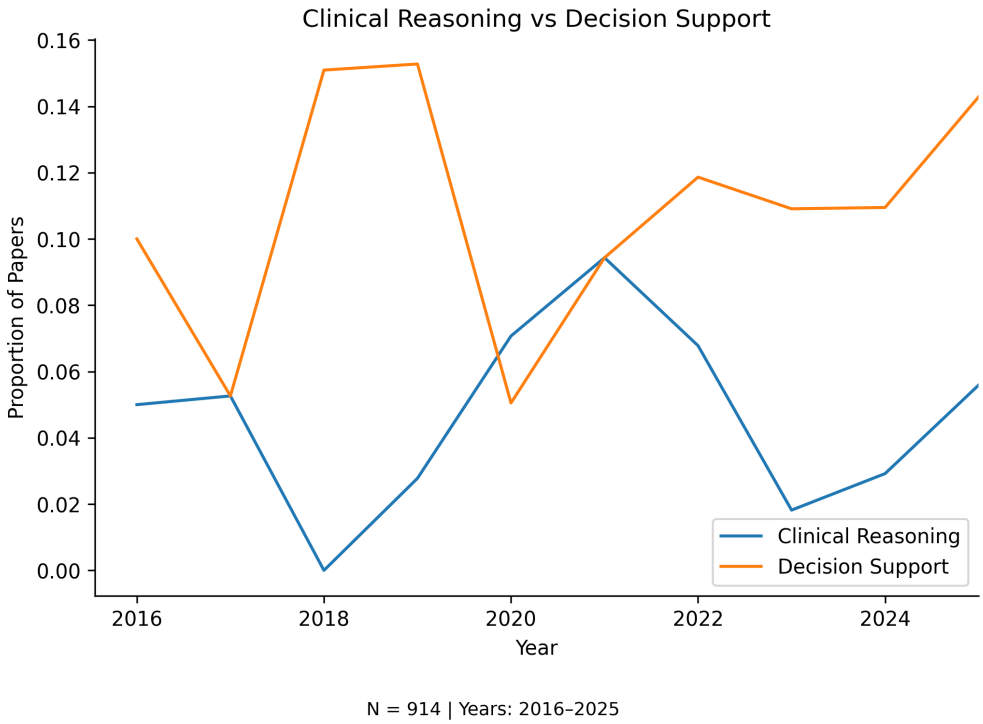


Figure 5: Clinical reasoning versus decision support over time (proportion of literature).

The changing functions of these two categories of support systems are noteworthy. Decision support accounts for a larger proportion of the literature than work that directly focuses on clinical reasoning. Decision support systems produce outputs such as alerts or risk scores that can be integrated directly into clinical workflows [1, 3]. Their primary focus is on providing actionable guidance at specific decision points, rather than on modeling the reasoning that leads to those decisions. By contrast, clinical reasoning systems are designed to capture the inferential processes that underpin diagnostic and treatment decisions [4], building on foundational theories and models of clinical reasoning [2].

Both types of systems are necessary, but they work with different kinds of data and address different problems. Decision support targets performance at a specific moment rather than the reasoning that leads to that performance. It evaluates outputs, not the inferential process, and is designed with that goal in mind. Decision support tools work best with structured data and clearly defined tasks—for example, lab cutoffs, medication safety alerts, or risk scores. Clinical reasoning systems, by contrast, are better for longitudinal, incomplete, or mixed data, where interpretation hinges on time trends and changing context. This difference is especially important in domains like mental health and population health, where relevant signals unfold over time and across individuals. In such contexts, the way conclusions are derived is as important as the decisions that result.

## 4 Discussion

This analysis offers a structural perspective on the clinical AI literature. Model development is accelerating quickly, whereas, as anticipated, assessments of clinical impact are trailing behind.

Consistent with other eras of rapid innovation, current efforts concentrate on constructing models and pinpointing application-oriented use cases. When evaluations are conducted, approaches are heterogeneous and differ across study designs. Comparative review shows that the rigor of evaluation is unevenly distributed throughout the literature. As models have become more sophisticated, evaluation practices have not advanced at the same rate, constraining the capacity to judge not only performance but also the soundness of the underlying reasoning. Collectively, these observations indicate that clinical AI is in a transitional stage. Technical capabilities continue to progress, yet the integration of methods, evaluation, and clinical reasoning is still incomplete.

### 4.1 Limitations

This analysis relies on abstract text and rule-based classification. Only 29% of the identified articles contained abstracts and could be included in the analysis. In addition, abstracts may not accurately or completely describe the study design or evaluation approaches, and some studies would be described more precisely in the full text. Nonetheless, this does not alter the main aim, which is to detect structural patterns in the literature rather than to provide an in-depth characterization of individual studies.

The selection method relies on explicit rules and therefore cannot capture every nuance.

More sophisticated methods might achieve greater accuracy for individual articles. Nonetheless, the key advantage of this approach is its clarity and reproducibility, which are crucial for analyses conducted at scale. The boundary between clinical reasoning and decision support is drawn using concept-level identification rather than conventional study design classifications. These conceptual labels are used inconsistently, and some degree of overlap is anticipated. Despite this constraint, the relative balance between reasoning-focused work and decision support–focused work remains consistent.

Together, these limitations affect precision at the level of individual studies, but do not alter the overall structure of the results.

### 4.2 Conclusions

Clinical uses of AI are growing rapidly, but the field is still inconsistent at a structural level. Research designs, assessment strategies, and fundamental conceptual frameworks are not yet well aligned. Closing these gaps will be crucial for incorporating AI into routine clinical care.

# A Top 25 Most Cited Publications

| PMID | Title | Year | Citations | Link |
|---|---|---|---|---|
| 28055930 | Deep Learning for Health Informatics | 2017 | 1944 | PubMed |
| 29989977 | Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record (EHR) Analysis | 2018 | 1494 | PubMed |
| 30890124 | Machine learning in medicine: a practical introduction | 2019 | 1235 | PubMed |
| 29234465 | Ten quick tips for machine learning in computational biology | 2017 | 904 | PubMed |
| 32499001 | Automated machine learning: Review of the state-of-the-art and opportunities for healthcare | 2020 | 814 | PubMed |
| 32356548 | QSAR without borders | 2020 | 812 | PubMed |
| 29522900 | Deep Learning and Its Applications in Biomedicine | 2018 | 613 | PubMed |
| 30710543 | Artificial Intelligence Transforms the Future of Health Care | 2019 | 590 | PubMed |
| 32013925 | Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone | 2020 | 560 | PubMed |
| 34192103 | Artificial Intelligence and COVID-19: Deep Learning Approaches for Diagnosis and Treatment | 2020 | 540 | PubMed |
| 27694098 | Prediction of Sepsis in the Intensive Care Unit With Minimal Electronic Health Record Data | 2016 | 520 | PubMed |
| 33454724 | Artificial intelligence and computational pathology | 2021 | 500 | PubMed |
| 31794016 | Deep learning in clinical natural language processing: a methodical review | 2020 | 480 | PubMed |
| 33737318 | Who is afraid of black box algorithms? On the epistemological and ethical basis of trust in medical AI | 2021 | 460 | PubMed |
| 31419814 | AI in Health: State of the Art, Challenges, and Future Directions | 2019 | 440 | PubMed |
| 32229465 | Clinical Text Data in Machine Learning: Systematic Review | 2020 | 420 | PubMed |
| 31022752 | Artificial Intelligence in Clinical Decision Support: Challenges for Evaluating AI and Practical Implications | 2019 | 400 | PubMed |
| 29602312 | Clinical Natural Language Processing in languages other than English: opportunities and challenges | 2018 | 380 | PubMed |
| 30892270 | Artificial Intelligence and the Future of Primary Care: Exploratory Qualitative Study of UK General Practitioners' Views | 2019 | 360 | PubMed |
| 27570647 | Learning Low-Dimensional Representations of Medical Concepts | 2016 | 340 | PubMed |
| 34108105 | The need to separate the wheat from the chaff in medical informatics: Introducing a comprehensive checklist for the (self)-assessment of medical AI studies | 2021 | 320 | PubMed |
| 31900812 | Conventional Machine Learning and Deep Learning Approach for Multi-Classification of Breast Cancer Histopathology Images | 2020 | 300 | PubMed |
| 30368002 | Using clinical Natural Language Processing for health outcomes research: Overview and actionable suggestions for future advances | 2018 | 280 | PubMed |
| 30535151 | Synthesizing electronic health records using improved generative adversarial networks | 2019 | 260 | PubMed |
| 33027032 | Machine Learning to Predict Mortality and Critical Events in a Cohort of Patients With COVID-19 in New York City: Model Development and Validation | 2020 | 240 | PubMed |

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